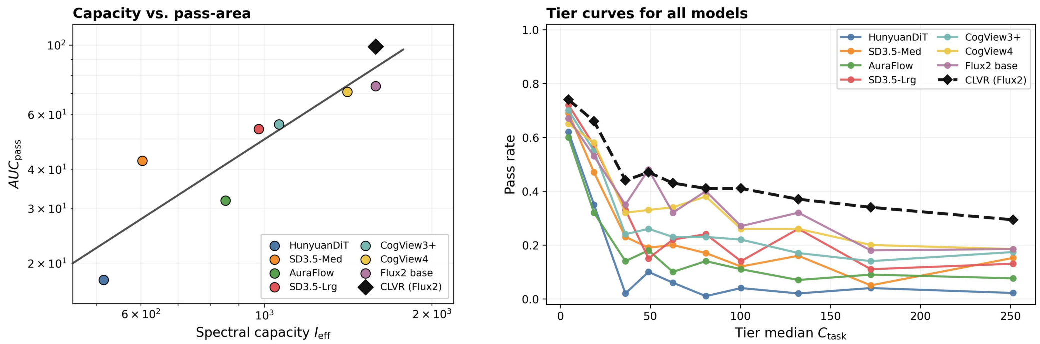

Figure 5: Quantitative results of the semantic complexity probe. C_{task} stratifies prompt difficulty (entities, relations, and hard constraints); I_{eff} is an entropy effective-rank spectral proxy for backbone capacity [35]. The left plot shows that CLVR achieves a higher Area Under the Pass-Complexity Curve (AUC Pass) compared to single-step models of similar capacity. The right plot illustrates that while single-step models' performance drops sharply as task complexity increases, CLVR maintains a high pass rate across all tiers.

high-complexity tiers. Compared to FLUX.2, CLVR improves AUC_{pass}, mitigating the structural capacity ceiling without expanding the DiT backbone.

4.4 Ablation studies

Prompt Rewrite vs. CLVR A natural question is whether the performance gains merely stem from the VLM rewriting the prompt into a more descriptive format, leveraging its inherent superior semantic understanding. In Table 4 (Panel A, B), we compare the FLUX.2 4B Distill baseline with an open-loop variant that uses Qwen3-VL 8B for prompt rewriting.

Open-loop prompt rewriting improves WiseBench scores (0.48 to 0.64) by providing external knowledge, but degrades GenEval scores (0.81 to 0.78) where instructions are already explicit. In contrast, CLVR achieves superior scores on both benchmarks (0.87 on GenEval, 0.74 on WiseBench). This confirms that CLVR's gains extend beyond the VLM's semantic enrichment. Instead, the improvements stem fundamentally from (1) task decomposition, which avoids forcing the model into one-shot generation and raises its capacity ceiling, and (2) iterative visual self-correction, which ensures each generative step remains within the diffusion model's reliable execution boundaries. This explains why our method comprehensively outperforms open-loop rewriting.

Training Alignment and Deployment Acceleration We ablate the training and deployment mechanisms to validate their individual contributions (Table 4, Panel A). Starting from the FLUX.2 4B base (0.74 on GenEval), applying CLVR via SFT alone improves the score to 0.78. Further applying PPRL boosts the score to 0.83, validating that proxy prompts effectively translate long-context histories into explicit reward signals for stable alignment.

Finally, employing DSWM yields a GenEval score of 0.87, surpassing both the RL-only (0.83) and distill-only (0.81) baselines. This confirms that alignment and distillation priors merge compatibly without destructive interference. By reusing the distillation prior, DSWM reduces iterative denoising from 28×2 to just 4 NFEs (number of function evaluations) per step, effectively resolving the computational bottleneck of closed-loop reasoning. We also provide an end-to-end accelerate analysis in Appendix A.8.

Qualitative Cases and Visual Comparisons To intuitively illustrate the mechanics of our framework, we provide detailed real-world Closed-Loop Visual Reasoning (CLVR) trajectories in the Appendix, Figure 7, showcasing how a complex prompt is systematically executed across multiple reasoning and generation steps. Furthermore, Figure 4 presents qualitative comparisons between our method (CLVR 4B) and strong baselines (e.g., FLUX.2 4B, Uni-CoT, Nano Banana 2 [34], and Qwen-Image) on challenging prompts. The results show that our framework approaches state-of-the-art proprietary instruction following and improves over open-source baselines.